Introduction to Software Engineering

Project Proposal & Plan

The student team is required to complete the **Project Proposal & Plan** documentation for the assigned course project, following the attached template.



Software Engineering Department
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Project Proposal & Plan

Objectives

This document focus on the following topics:

- ✔ Completing the Project Proposal & Plan document with the following sections:
  - Vision
  - Proposed Solution
  - Project Plan
  - Risk Management
  - Tools & Collaboration Setup
- ✔ Including an AI usage log for this phase
- ✔ Understanding the Project Proposal & Plan document.

1 Member Contribution Assessment

Planned PA1 contribution is allocated equally because each member owns a major assessed section. The PM will replace or supplement the Jira references with final URLs, commits, drafts and review evidence before submission; totals must remain exactly 100%.

Student ID	Name	Role(s)	Contribution	Evidence / accountable deliverable
24127169	Truong Le Trung Hieu	PM; Proposed Solution; Integrator	25%	Jira DID-7 and DID-14; PA1 Proposed Solution and integration review
24127130	Tran Minh Tien	Business Analyst; Document coordinator	25%	Jira DID-6 and DID-13; Vision and Scope; contribution and AI log consolidation
24127020	Nguyen Thanh Dat	Designer; Project Plan lead	25%	Jira DID-8, DID-9, DID-10, DID-17, DID-19; WBS, estimates, schedule and cost
24127035	Vo Dinh Cao Minh Hao	Tester; collaboration compliance	25%	Jira DID-11, DID-12 and DID-18; risk register, tools audit and independent QA

2 Vision

2.1

2.2 Preliminary Problem Statement

Ho Chi Minh City residents must often make short-notice decisions about travel and outdoor activity while air-quality and weather conditions change. Fine particulate matter can increase health risk, especially for children, older adults and people with respiratory conditions. Rain also affects exposure, visibility and travel planning. Relevant information is fragmented across station-data portals, model-based services and weather applications, so users must interpret sources with different coverage, timestamps and uncertainty.

DiDung addresses this problem with a responsive web application that brings PM2.5-derived air-quality information, weather context, short-term forecasts and personal in-app alerts into one place. It is intended for a modern HTML5 browser on desktop or mobile and a cloud-hosted backend. The product displays the source and update time of each value so measured observations are not confused with model estimates.

The product is needed because users currently spend time switching between sources and may not know whether a value is measured, modeled or stale. DiDung will support a small, clearly declared set of locations in Ho Chi Minh City, use scheduled collection rather than claim real-time sensing, and provide simple decision support rather than medical advice. Its value is a consistent view and threshold-based warning, not a promise of street-level precision.

Evidence and assumptions. The approved PA0 identifies PM2.5, rain and high-tide flooding as the original problem space. Repository guidance subsequently narrows the implementable core to air quality with weather as supporting context. OpenAQ station coverage, data gaps and update delay must be measured during requirements work; Open-Meteo values are model outputs and may not support ward-level claims. A reliable free tide source has not yet been validated. These uncertainties are treated as risks and scope gates, not as confirmed capabilities.

2.3 Product Position Statement

Element	Content
For	Residents, commuters, outdoor workers and people sensitive to poor air quality in Ho Chi Minh City
Who	Need a quick, understandable view of current conditions and the next few hours before going outside
The DiDung	Responsive environmental-condition monitoring and alert web application
That	Combines sourced PM2.5/AQI information, weather context, forecasts and personal in-app thresholds
Unlike	Checking separate weather, air-quality and news sources without a shared timestamp or personalized warning
Our product	Labels provenance and freshness, distinguishes measured from modeled data, and stays within a small HCMC study area

2.4 Stakeholders and Users

#	Stakeholder / user	Description	Main interests / influence
1	Residents and commuters	Primary end users planning daily travel	Fast overview, freshness, understandable thresholds; high interest
2	Residents in flood-prone areas	Conditional user group from PA0	Area-level warning only if tide/flood data is validated
3	Outdoor workers and athletes	Users exposed for long periods	Short-term outlook and mobile-browser usability
4	Sensitive groups and families	Users who may choose lower thresholds	Personal settings, clear limitations; no medical advice
5	System administrator	Operates data pipeline and service	Fetch status, data gaps, error logs and deployment health
6	Data providers	OpenAQ, Open-Meteo; tide source if validated	Attribution, API conditions, quotas and responsible usage
7	Teaching staff and TA	Sponsor/reviewer for coursework	Traceable scope, evidence, process compliance and feasible delivery

2.5 Scope and Key Features

MoSCoW priorities below reconcile the approved PA0 feature list with the three-week coding window. Must-have items define PA5 acceptance. Should/Could items are implemented only after Must-have exit criteria pass.

ID	Feature	Description	Priority
F01	Data collection and storage	Scheduled PM2.5/weather ingestion, validation, provenance and timestamps	Must
F02	Short-term forecast	1–6 hour PM2.5 forecast using provider data or a validated baseline	Must
F03	Baseline and optional ML comparison	Evaluate a simple model against persistence; use only if better	Should

ID	Feature	Description	Priority
F04	Tide and flood-risk warning	Area-level rule and known hotspots only when a source is validated	Won't for MVP
F05	Accounts and regional alerts	Register/login, select supported location and personal thresholds	Must
F06	Alert banner and notification center	Create and retain in-app alerts when a forecast crosses a rule	Must
F07	Dashboard	Sourced map plus history and forecast charts for supported points	Must
F08	General advice	Short non-medical guidance derived from displayed risk levels	Could
F09	Admin monitoring	Pipeline runs, stale-data warnings and forecast-error summaries	Should
F10	Mobile push notifications	Native/web push beyond the in-app notification center	Won't

Scope boundaries and constraints

- ✔ Out of scope: native mobile application and push alerts; route navigation; medical advice; coverage outside the declared HCMC study area; unsupported ward-level AQI; verified street-level flood prediction; and user flood reports. Tide/flood functionality remains excluded from the MVP until a reliable source and validation rule are approved.

Key constraints: external-source availability and quotas; sparse or delayed station measurements; model-data resolution; hourly collection targets subject to provider refresh rates; four-member capacity; and deployment free-tier limits. The interface should work in current mobile/desktop browsers, show a useful dashboard response within about 3 seconds under coursework demo load, and provide source, timestamp and missing-data states. These are planning targets to validate, not measured results.

3 Proposed Solution

3.1 Preliminary Architecture

C4 System Context. DiDung is the system boundary. People use the responsive web application; administrators monitor ingestion and errors. External providers supply observations or model forecasts. A tide/flood input is shown as conditional because no reliable source has yet been validated.

System boundary: DiDung web app and backend — API, scheduled ingestion, database, forecast/baseline evaluation, alert rules, dashboard and notification center.

End-to-end flow: (1) the scheduler requests external data; (2) the backend records raw provenance, normalizes units/time and flags gaps; (3) validated values and forecast runs are stored; (4) the baseline or approved model produces displayable outlooks; (5) threshold rules create de-duplicated in-app notifications; and (6) the web UI shows maps/charts with source, freshness and uncertainty. If data is insufficient, the system displays the latest valid value and a stale/unavailable state instead of fabricating a forecast.

Person / system	Relationship with DiDung	Status
Resident / commuter / outdoor user	Views sourced conditions and forecasts; manages thresholds; receives in-app alerts	In scope
System administrator	Reviews pipeline runs, stale data and deployment health	Should-have
OpenAQ	Provides available station-based PM2.5 observations	Proposed; coverage to verify
Open-Meteo Weather / Air Quality	Provides weather and model-based forecast/context data	Proposed; label as modeled
Published tide/flood source	Would provide tide tables or reviewed hotspot input	Conditional; not MVP until validated

3.2 Technology Stack

Layer	Technology	Reason for choice and decision status
Front end	Next.js + Shadcn/ui	Responsive UI; map and chart libraries fit dashboard needs. React is proposed pending team confirmation.
Back end	Next.js	Small typed API, compatible with data processing. Confirmed only after a technical spike.
Database	Supabase (managed PostgreSQL)	Relational users/alerts plus timestamped readings and forecast runs.

Layer	Technology	Reason for choice and decision status
Scheduler / processing	Supabase pg_cron	Simple hourly jobs and reproducible baseline evaluation; avoid microservices.
Hosting / deployment	Vercel (Next.js) + Supabase Cloud	Repeatable environment; provider/free-tier choice remains to be validated.
Testing & CI	Vitest, Playwright candidate, GitHub Actions planned	Unit/API/UI checks and PR gate; CI is planned, not yet evidenced as configured.

3.3 Hardware and Operating Environment

Development requires each member's laptop, Git, a current browser, Python/Node tooling and Docker where supported. Test data and secrets remain outside Git. The proposed runtime is one containerized web/API service, one scheduled worker and PostgreSQL; a separate GPU is not required because the baseline and small tabular models run on CPU.

Environments: local development uses sample or limited API data; test uses seeded fixtures and mocked provider failures; production/demo uses restricted real credentials and scheduled collection. Configuration is supplied through environment variables. Logs record source, fetch time, result count and failure reason without storing passwords or API keys.

Operational controls: cache or batch requests within documented quotas, retry transient failures with limits, retain old forecast runs for evaluation, identify measured versus modeled values, and display missing/stale states. The baseline is the release fallback; an ML model is promoted only after time-based evaluation shows an improvement on held-out data.

4 Project Plan

4.1 Process and Phase Gates

The project uses course phase gates with controlled iteration inside each phase. Each PA becomes a reviewed baseline only after its exit criteria pass; later changes must be recorded and traced. Dates below are proposed planning windows, not verified Moodle deadlines or evidence of completion. The PM confirms official dates and records actual status. Weekly team reviews are on Tuesday (time to be agreed), with internal handoff targeted at least 24 hours before each verified submission deadline.

PA	Phase	Planning date	Observable exit criteria
PA1	Initiation & planning	W4, 19–25 Oct 2026; proposed	Proposal source/PDF assembled; ownership, WBS, estimates, risks and QA evidence reviewed
PA2	Requirements	End of week 5; confirm Moodle	SRS/use cases/acceptance criteria reviewed; data feasibility report and scope decision recorded; approval pending until evidenced
PA3	Design	End of week 7; confirm Moodle	C4/UML/data/UI designs trace to Must-haves; design review defects resolved
PA4	Test planning	End of week 8; confirm Moodle	Test plan/cases/data/environment reviewed; requirements traceability complete
PA5	Implementation, test & deployment	End of week 10; confirm Moodle	Must-haves deployed; tests pass; known limitations and evidence packaged
Final	Presentation & demo	Week 11; confirm Moodle	Stable demo, slides, backup evidence and speaking roles rehearsed

4.2 Work Breakdown Structure (WBS)

The deliverable-oriented WBS covers PA1 through final presentation. Level 1 groups deliverables, level 2 identifies components, and level 3 contains assignable work packages. Only level-3 leaves carry effort; parent rows are not additional work. PA1 includes all proposal sections and submission QA. Jira issues should use these IDs or link back to them. Developer assignments remain TBD; the PM/team must name accountable owners by the PA2 gate.

WBS	Deliverable / work package	Main / backup	Acceptance evidence	Scope trace
1	Project initiation and control	Hiếu / Tiến	PA1 baseline and weekly control records	All
1.1	Proposal and scope	Tiến / Hiếu	Vision, stakeholders, MoSCoW reviewed	F01–F10

WBS	Deliverable / work package	Main / backup	Acceptance evidence	Scope trace
1.1.1	Vision, scope and Proposed Solution baseline	Tiến / Hiếu	Vision, stakeholders, scope and technical proposal reviewed	F01–F10
1.2	Planning and governance	Đạt / Hiếu	WBS, estimates, schedule, cost and risks	All
1.2.1	PA1 planning, integration and QA	Đạt / Hiếu; QA Hào	Plan, risks, tools, contribution assessment and truthful AI log; source/PDF QA closure	All
2	Requirements baseline	Tiến / Hiếu	Approved SRS and traceability	Must-haves
2.1	User and system requirements	Tiến / Hào	Use cases and measurable acceptance criteria	F01,F02,F05–F07
2.1.1	Data-source feasibility spike	TBD developer / Tiến	Coverage/gap report and source decision	F01,F02,F04
2.2	Functional requirements	Tiến / Đạt	Reviewed requirements and acceptance criteria	F05–F07
2.2.1	Alerts and dashboard requirements	Tiến / Đạt	Reviewed flows and edge cases	F05–F07
3	Solution design	Đạt / Hiếu	Design package reviewed	Must-haves
3.1	Technical design	Hiếu / TBD developer	Architecture, data and API review	F01,F02,F05,F06
3.1.1	Architecture, data model and API design	Hiếu / TBD developer	C4/data/API diagrams trace to requirements	F01,F02,F05,F06
3.2	Interface design	Đạt / Tiến	Reviewed UI prototype	F07
3.2.1	Responsive UI and visualization design	Đạt / Tiến	Prototype and usability review	F07
4	Test planning	Hào / Tiến	PA4 test baseline	Must-haves
4.1	Verification preparation	Hào / Tiến	Test plan and traceability	Must-haves
4.1.1	Test cases, fixtures and traceability	Hào / Tiến	Reviewed cases including failure/stale data	F01,F02,F05–F07
5	Implementation and integration	TBD developer / Hiếu	Deployed increment and PR evidence	Must-haves
5.1	Data pipeline	TBD developer / Hào	Ingestion and storage checks	F01
5.1.1	Ingestion, normalization and storage	TBD developer / Hào	Automated checks and source metadata	F01
5.2	Forecast	TBD developer / Hào	Baseline evaluation	F02; F03 optional
5.2.1	Forecast baseline and evaluation	TBD developer / Hào	Held-out metrics versus persistence	F02,F03
5.3	Accounts and alerts	TBD developer / Tiến	Acceptance test evidence	F05,F06

WBS	Deliverable / work package	Main / backup	Acceptance evidence	Scope trace
5.3.1	Accounts, alerts and notification center	TBD developer / Tiến	Acceptance tests pass	F05,F06
5.4	Dashboard	TBD developer / Đạt	UI and provenance checks	F07
5.4.1	Dashboard map and charts	TBD developer / Đạt	Responsive UI tests and provenance labels	F07
5.5	Release verification	Hào / Hiếu	Regression and deployment evidence	Must-haves
5.5.1	System test, fixes and deployment	Hào / Hiếu	Regression report, deployed URL and limitations	Must-haves
6	Final presentation	Hiếu / Đạt	Demo and presentation package	Delivered scope
6.1	Presentation preparation	Hiếu / Đạt	Slides, rehearsal and backup	Delivered scope
6.1.1	Final demo and presentation	Hiếu / Đạt	Rehearsed demo, slides and backup recording	Delivered scope

4.3 Estimation

Effort is estimated in person-hours using expert judgement and analogy with PA0. Uncertain packages retain optimistic/most-likely/pessimistic ranges as risk checks; the baseline uses the most-likely values, not weighted PERT means. The 13 leaves sum to 260 person-hours, not 280. Duration follows dependencies and available capacity. Assuming four members, W4–W7 and W11 at 5–8 h/member/week plus W8–W10 at 8–12 h gives 196–304 h. The 260 h baseline fits only above minimum availability; confirm individual skills and commitments after assigning Developer TBD. Record actual PA1 hours and rebase remaining effort before committing.

WBS ID	Work package	Owner	Effort (h)	Method / assumptions
1.1.1	Vision, scope and Proposed Solution	Tiến	10	Expert judgement; includes review and corrections
1.2.1	PA1 planning, integration and QA	Đạt/Hiếu/Hào	12	Analogy with PA0 plus render/recheck
2.1.1	Data-source feasibility spike	TBD developer/Tiến	14	O/M/P 10/14/22; baseline M=14; coverage risk
2.2.1	Alerts/dashboard requirements	Tiến/Đạt	14	Expert judgement; use cases + acceptance criteria
3.1.1	Architecture, data model, API	Hiếu/TBD	18	Expert judgement; includes design review
3.2.1	Responsive UI design	Đạt	14	Analogy; prototype plus review
4.1.1	Test cases and fixtures	Hào	18	Requirements-based estimation
5.1.1	Ingestion and storage	TBD developer	32	O/M/P 24/32/48; baseline M=32
5.2.1	Forecast baseline/evaluation	TBD developer/Hào	28	Judgement; data quality drives variance; ML optional

WBS ID	Work package	Owner	Effort (h)	Method / assumptions
5.3.1	Accounts and alerts	TBD developer	26	Analogy with standard auth/rule workflow
5.4.1	Dashboard map and charts	TBD developer/Đạt	30	Expert judgement; responsive UI included
5.5.1	Integration, system test, deployment	Hào/Hiếu	32	Includes regression and contingency fixes
6.1.1	Final demo and presentation	Hiếu/Đạt	12	Expert judgement; rehearsal + backup

4.4 Schedule

Planning basis. Weeks and gates below are proposed; Hiếu must confirm Moodle deadlines and record actual progress. CP = critical activity; ■ = planned work; ◆ = proposed gate. W4–W11 is the active planning window; no completion status is implied.

Planned timeline

Activity	W4 19–25 Oct	W5 26 Oct–1 Nov	W6 2–8 Nov	W7 9–15 Nov	W8 16–22 Nov	W9 23–29 Nov	W10 30 Nov–6 Dec	W11 7–13 Dec
A PA1 baseline · CP	■◆							
B Requirements/data spike · CP		■◆						
C Technical contracts · CP			■					
UI design			■	■◆				
H Test plan and fixtures			■	■				
D Ingestion and storage				■	■			
E Baseline forecast						■		
F Accounts/alerts + dashboard · CP				■	■	■	■	
Integration, test and deployment						■	■◆	
J Final demo · CP								■◆

Gate I is the end-W10 release decision shown by ◆ in the integration row. It passes only after E, F and H are complete and the full 32 h of integration, testing and deployment are finished. UI design continues into W7; early implementation uses the reviewed W6 contracts and mocks.

Dependencies and critical path

ID	Predecessor	Timing*	Total float
A	None	0–1 (W4)	0
B	A	1–2 (W5)	0
C	B	2–3 (W6)	0
D	C	3–5 (W7–W8)	1 week
E	D	5–6 (W9)	1 week
F	C	3–7 (W7–W10)	0
H	B	2–4 (W6–W7)	3 weeks
I	E, F, H + verification complete	At 7 (end W10)	0
J	I	7–8 (W11)	0

*Timing is elapsed planning weeks from the start of W4; I is a zero-duration release gate. Dependencies are finish-to-start, except that integration checks completed interfaces incrementally before the final release gate.

Critical path: A → B → C → F → I → J = 8 weeks.

Float is schedule flexibility, not spare labor. This dependency model remains conditional on named developers and confirmed availability; recalculate it after resource allocation.

Weekly workload and capacity

Week	Work allocation (person-hours)	Total h	Capacity h
W4	PA1 22	22	20–32
W5	Requirements/data spike 28	28	20–32
W6	Technical design 18; test preparation 14	32	20–32
W7	UI 14; tests 4; ingestion 8; accounts 2; dashboard 4	32	20–32
W8	Ingestion 24; accounts 10; dashboard 10	44	32–48
W9	Forecast 28; accounts 8; dashboard 2; early integration 6	44	32–48
W10	Accounts 6; dashboard 14; final integration/test/deploy 26	46	32–48
W11	Demo and presentation 12	12	20–32
Total	Baseline person-hours	260	196–304

Implementation effort is spread across W7–W10 as 14 / 44 / 38 / 20 h. Early integration in W9 checks completed interfaces; final wiring and regression follow finished features. The full 32 h verification package is included in the 260 h baseline.

Capacity check. W6–W7 use the 32 h/week upper limit; W10 uses 46/48 h. Assign and level Developer TBD work by PA2, confirm individual commitments and replan if availability is lower. The 24-hour review target is not a one-week buffer. Effort and calendar duration are different measures.

4.5 Roles and Responsibilities

PA1 author–review–integrate protocol: each Jira assignee is the accountable author; a different member reviews the section; Hiếu integrates; Hào performs independent rubric-based QA. Weekly meetings are on Tuesday; time is to be agreed. Owners hand off source and evidence at least 24 hours before the verified official deadline; defects record severity, owner and resolution target. Developer ownership remains TBD and must be confirmed by the PM/team by PA2 before implementation is committed.

Role	Main	Backup	Responsibilities
Team lead / Project manager	Truong Le Trung Hieu	Tran Minh Tien	Plan, dependencies, TA contact, integration and submission evidence
Business analyst	Tran Minh Tien	Truong Le Trung Hieu	Scope, requirements, acceptance criteria and documentation consistency
Designer / Project Plan lead	Nguyen Thanh Dat	Truong Le Trung Hieu	UI design; WBS, estimates, Gantt and cost consistency; architecture presentation
Developers	TBD — PM approval required	TBD; confirm by PA2	Implement, unit test and review PRs; named owners and weekly availability required before committing implementation
Tester	Vo Dinh Cao Minh Hao	Tran Minh Tien	Test planning, system tests, defect tracking and independent PA QA

4.6 Human Resources and Cost Estimate

This is a reproducible coursework estimate, not an actual payment. A blended hypothetical rate of VND 120,000 per person-hour is used for all student roles. Planned labor is 260 hours × VND 120,000 = VND 31,200,000. A separate 15% monetary contingency is VND 4,680,000; planning total is VND 35,880,000. Contingency does not add hours or scope to the 260 h baseline.

Other resources and assumptions: members use existing laptops and internet connections (no incremental project cost); GitHub/Jira/Discord and candidate hosting/database/API services are assumed to use educational or free tiers during the course. No provider price is claimed as

verified. If a quota or free tier is insufficient, the team reduces refresh frequency/demo load or seeks PM/TA approval before incurring cost. Effort and cost must be refreshed after the data-source spike and connected to the final Gantt.

Work area	Hours	Rate (VND/h)	Labor cost (VND)
Initiation, requirements and planning	50	120,000	6,000,000
Architecture and UI design	32	120,000	3,840,000
Test planning and fixtures	18	120,000	2,160,000
Implementation work packages	116	120,000	13,920,000
Integration, test and deployment	32	120,000	3,840,000
Final demo and presentation (6.1.1)	12	120,000	1,440,000
Labor subtotal	260		31,200,000
Contingency (15%)			4,680,000
Planning total			35,880,000

5 Risk Management

Internal convention: Probability P and Loss/Impact L use a 1–5 scale; Risk Exposure RE = P × L. For team prioritization, RE 16–25 is High, 8–15 Medium, and 1–7 Low. This scale is the team's working convention, not a lecturer-provided rule. The Type column follows Sommerville's project/product/business categories (2011, Ch. 22). Scores, thresholds and triggers are planning assumptions to validate. The team reviews the register at Thu/Sun checkpoints and when triggers occur, revising scores and responses as evidence changes.

ID	Risk	Type	P	L	RE / level	Mitigation / trigger / contingency	Owner
R1	OpenAQ has sparse, delayed or gapped HCMC station data	Product	4	5	20 / High	Trigger: <3 usable points or >20% missing. Mitigation: measure coverage early and retain provenance. Contingency: use a smaller study area and separately labeled model data.	Tiến
R2	Modeled air-quality resolution is presented as local measurement	Product	3	5	15 / Med.	Trigger: UI or report labels modeled data as local measurement or makes unsupported ward-level claims. Mitigation: label source and resolution. Contingency: limit the map to supported points and add limitation text.	Hiếu
R3	No reliable free tide/flood source is validated	Product	4	4	16 / High	Trigger: the source spike ends without a stable source, usable coverage or an approved validation rule. Mitigation: time-box the spike and use an approval gate. Contingency: exclude F04 from the MVP and retain it as future work.	Tiến
R4	Insufficient history makes forecast model unreliable	Product	4	4	16 / High	Trigger: history is insufficient for a time-based holdout or the model underperforms persistence. Mitigation: profile history and evaluate against the persistence baseline. Contingency: publish the provider/baseline forecast with uncertainty; do not promote ML.	Hào

ID	Risk	Type	P	L	RE / level	Mitigation / trigger / contingency	Owner
R5	External API quota, schema change or outage breaks ingestion	Project / product	3	4	12 / Med.	Trigger: quota warning, schema-contract failure, repeated provider error or stale-data alarm. Mitigation: use contract tests, bounded retries and caching. Contingency: use fixtures or the last valid value and show unavailable status.	Hiếu
R6	Unconfirmed developer ownership delays implementation	Project	4	5	20 / High	Trigger: no named main/backup by PA2 or confirmed capacity falls below the plan. Mitigation: PM assigns main/backup after the technical spike and reviews load weekly. Contingency: reallocate work and cut Should/Could scope.	Hiếu
R7	Integration/document handoffs miss internal deadlines	Project	3	4	12 / Med.	Trigger: source/evidence misses a Thu/Sun checkpoint or a critical defect remains open inside the 24-hour buffer. Mitigation: use evidence-based exit criteria. Contingency: freeze scope and escalate unresolved critical defects.	Đạt
R8	Secrets or unreviewed changes enter repository	Project / product	2	5	10 / Med.	Trigger: direct push to main, missing PR review, secret-scan alert or a tracked .env file. Planned mitigation: protected main, peer review, .gitignore and secret scanning. Contingency: revoke or rotate secrets and revert through a reviewed PR.	Hào

6 Tools & Collaboration Setup

Evidence status for PA1: this section separates proposed controls from verified configuration. No Jira, GitHub, CI or Discord item is treated as complete until a URL, screenshot, access record or workflow run is attached. The draft references DID-6 through DID-19, but the board URL, owners, due dates, priorities and statuses remain evidence pending.

- ✔ Moodle: the official deadline, required filenames and submission procedure are not evidenced here. The PM must verify and record them before submission.
- ✔ Communication: Discord is the intended team channel. Evidence that all members and the TA can access it remains pending.
- ✔ Task board: Jira is the planned board. Each WBS leaf needs an owner, due date, priority, status, acceptance evidence and a link to its WBS ID. Do not mark this check complete until the board URL and field values are attached.
- ✔ Repository audit: attach the GitHub URL, then verify README, .gitignore, /pa/PA1, /docs/management/weekly and instructor/TA access. These items remain evidence pending in this section.
    - Planned branch and merge policy: use feature branches and Conventional Commits; merge through a pull request with at least one peer review. Main-branch protection remains unverified until settings evidence is attached.
    - Repository hygiene check: verify README and .gitignore from the linked repository before code or credentials are introduced. Never commit .env files, API keys or passwords; record remediation if a secret scan raises an alert.

- Commit-message check: inspect the linked Git history before rating compliance.

  Apply feat:, fix:, docs:, test: and chore: prefixes going forward; do not report compliance without the commit log or repository URL.

✔ Meetings and reports: internal Thu/Sun checkpoints are planned; exact times require team confirmation. Sunday 23:59 is a proposed weekly-report cutoff and must be confirmed by the team. Reports belong in /docs/management/weekly; record attendance, decisions, blockers and next owners. Do not claim a report exists without a file or commit link.

Repository structure and readiness

Planned CI/test approach: GitHub Actions with backend unit/API tests and selected browser smoke tests. CI remains planned until a workflow file and successful run URL are attached. PM follow-up items are Jira evidence, official Moodle deadline, main protection, .gitignore, repository access, developer ownership and CI evidence. Hào records PASS/NEEDS FIX with severity and rechecks critical defects before sign-off.

Folder	Content	Evidence status
/src	Source code	Planned; not verified
/docs/management	Project plan and /weekly reports	Required path; week-2 report evidence pending
/docs/requirements	Vision, SRS, use cases, mockups	Planned; not verified
/docs/design	C4, ADRs, UML and UI design	Planned; not verified
/docs/test	Test plan, cases and reports	Planned; not verified
/docs/ai-usage	AI usage logs	Required path; evidence pending
/pa/PA1 … /pa/PA5	PDF and editable source per PA	PA0 exists; PA1 folder planned

7 AI Usage Log

The entries below records the significant AI assistance used for this draft. The accountable student and section owners must independently verify the content, correct it using team evidence and be able to explain the final submission.

#	Member	Tool	Task / purpose	How the output was verified and changed
1	Vo Dinh Cao Minh Hao	ChatGPT / Codex	Initial drafting and consistency review of the PA1 risk, tools/collaboration and document-integration content	Checked against PA0, PA1 instructions and repository evidence; removed unsupported claims, marked unknown dates/configuration as pending, and requires owner/PM review before submission
2	Nguyen Thanh Dat (Project Plan lead)	ChatGPT / Codex	19 Oct 2026: Requested evaluation and revision suggestions for Section 4 Project Plan (4.1–4.6): phase gates, three-level WBS, effort/capacity, schedule/dependencies/ critical path, roles and cost; applied targeted corrections and improved Schedule formatting.	AI recalculated the leaf estimates as 260 h and aligned cost at VND 120,000/h; checked WBS parent levels and weekly totals; retained unverified deadlines/scope decisions as pending. Student/team verification of actual progress, availability, developer assignments and course requirements remains required.